

Conservation and Transport Equations

Control volumes, Gauss's theorem and Reynolds transport theorem

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What we will calculate

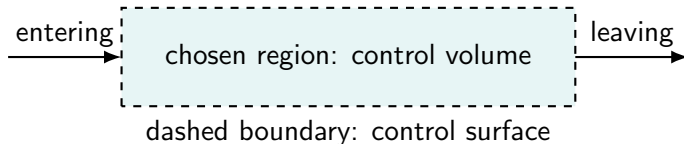
We choose a region, identify the quantity inside it, and calculate how that quantity changes.

1. Define the region and its boundary.
2. Calculate transport entering and leaving.
3. Include production, consumption or applied forces as appropriate.
4. Relate these terms to the change inside the region.

We will connect the balance over a finite region to the equation at each point. Each new symbol is defined where it first appears.

A finite control volume and its control surface

A **control volume (CV)** is a region chosen for the calculation. “Finite” means that it has a specified, nonzero size. A pipe section or a tank is an example. Its complete boundary is the **control surface (CS)**. Fluid may enter or leave across this surface. A fixed CV stays in the same position and has the same shape.



A material system follows the same matter

A **material system** contains the same particles at every time. Its boundary moves and deforms with those particles. A control volume is a chosen region; it can contain different particles at different times. For example, water passing through a fixed pipe section enters at one end and leaves at the other. The system and CV may occupy the same region at one instant. Later, they generally occupy different regions. Reynolds transport theorem connects the rate of change for the material system with measurements made in a CV.

Define the transported quantity before writing a balance

Let t be time (s), $\mathbf{x} = (x, y, z)$ position (m), and ρ total mass density (kg/m^3). Let ψ be a property per unit mass. Define $C = \rho\psi$, the amount of that property per unit volume.

Property	ψ	unit of ψ	unit of C
Total mass	1	dimensionless	kg/m^3
Species A mass	mass fraction w_A	kg A/kg mixture	$\text{kg A}/\text{m}^3$
Energy	specific energy e	J/kg	J/m^3
x-momentum	velocity u_x	m/s	$\text{kg}/(\text{m}^2 \text{ s})$

Mass fraction means species mass divided by total mixture mass.

Amount inside and rate of accumulation

Write V for the chosen region and dV for a small volume element (m^3). The total amount inside is

$$B_{CV}(t) = \int_V C(\mathbf{x}, t) dV.$$

B_{CV} has the unit of the property: kg for mass, J for energy, or kg m/s for momentum. The **rate of accumulation** is dB_{CV}/dt . It is positive when the amount increases and negative when it decreases. A decrease is also called depletion. For mass, amount and accumulation rate have different units: kg and kg/s.

Flow rate and flux are different quantities

- ▶ **Volumetric flow rate** is the volume of fluid passing through a cross-section per unit time. Its unit is m^3/s .
- ▶ **Mass flow rate** is the mass of fluid passing through a cross-section per unit time. Its unit is kg/s .
- ▶ **Flux** is a transfer rate per unit area. The vector $\mathbf{J}$ gives the amount transferred per unit area per unit time and the direction of transport.

Mass flux has unit $\text{kg}/(\text{m}^2 \text{ s})$; heat flux has unit W/m^2 ; momentum flux has unit N/m^2 .
Multiplying a uniform normal flux by area gives the transfer rate through that area.

Influx, efflux and their signs

Following Welty's control-volume terminology:

- ▶ **Influx**: flow entering the control volume.
- ▶ **Efflux**: flow leaving the control volume.
- ▶ **Net efflux**: outflow minus inflow.
- ▶ **Net influx**: inflow minus outflow.

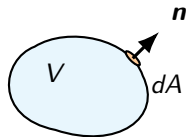
For mass flow these rates are in kg/s. If 2 kg/s enters and 3 kg/s leaves, net mass efflux is 1 kg/s. With no source, the mass inside decreases at 1 kg/s. Welty, Wicks, Wilson and Rorrer, *Fundamentals of Momentum, Heat, and Mass Transfer*, 5th ed., Chapter 4.

The outward normal defines the sign of surface transport

Let dA be a small surface area (m^2), and $\mathbf{n}$ its outward unit normal. $\mathbf{n}$ is perpendicular to the surface, points out of the CV, and is dimensionless. The dot product $\mathbf{J} \cdot \mathbf{n}$ is the signed outward flux. It is positive for outward transport and negative for inward transport.

$$\text{net outward transfer rate} = \oint_{\partial V} \mathbf{J} \cdot \mathbf{n} dA.$$

∂V means the entire boundary of V . The circle on $\oint$ indicates a closed surface. The integral adds the signed transfer rates over all its small patches.



Generation, consumption and the rate balance

For a **fixed control volume**, begin with

$$\text{accumulation} = \text{in} - \text{out} + \text{generation} - \text{consumption}.$$

The integral equation has the same form:

$$\frac{d}{dt} \int_V C dV = - \oint_{\partial V} \mathbf{J} \cdot \mathbf{n} dA + \int_V S dV.$$

The left side is accumulation. The surface integral is **out minus in**; its negative therefore gives **in minus out**.

S is generation minus consumption per unit volume per unit time: $\text{kg}/(\text{m}^3 \text{ s})$ for species mass, or W/m^3 for energy. Its volume integral gives total net production.

Every term is a rate. For total mass, $S = 0$ in ordinary nonnuclear processes. Reactions can change species masses while conserving total mass.

Examples of generation and consumption

Generation means production of the selected species inside the control volume. For example, a reaction that forms CO_2 generates CO_2 .

Consumption means removal of the selected species by a process inside the control volume. For example, combustion consumes oxygen.

Material entering or leaving through the boundary is **inflow or outflow**, not generation or consumption.

Example: species A in a reacting vessel

Let the vessel volume be $V = 2 \text{ m}^3$. Two reactions occur: one produces A at a uniform rate $r_A^+ = 0.03 \text{ kg}/(\text{m}^3 \text{ s})$, and another consumes A at $r_A^- = 0.01 \text{ kg}/(\text{m}^3 \text{ s})$.

$$S_A = r_A^+ - r_A^- = 0.02 \text{ kg}/(\text{m}^3 \text{ s})$$

Here S_A is the net production rate per unit volume. The total net production rate is

$$S_A V = 0.02(2) = 0.04 \text{ kg/s}.$$

In a closed vessel, A therefore accumulates at 0.04 kg/s while these rates remain constant. Chemical reactions change species amounts but conserve the total mass of all species.

Steady state does not mean that transport stops

At steady state, the fields in a fixed CV do not change with time. The accumulation term is zero. Mass may still enter and leave continuously. With no mass source, equal entering and leaving mass flow rates are required. A tank with 2 kg/s entering and 1 kg/s leaving is not steady: its mass increases at 1 kg/s. A rigid duct completely filled with constant-density water cannot undergo this increase while its volume and density remain fixed. Unequal imposed mass flow rates then contradict the assumptions.

Steady flow does not mean that flow has stopped

The rate balance is

$$\text{accumulation} = \text{in} - \text{out} + \text{generation} - \text{consumption}.$$

If

$$\text{accumulation} = 0, \quad \text{generation} - \text{consumption} = 0,$$

then

$$\boxed{\text{in} = \text{out}.}$$

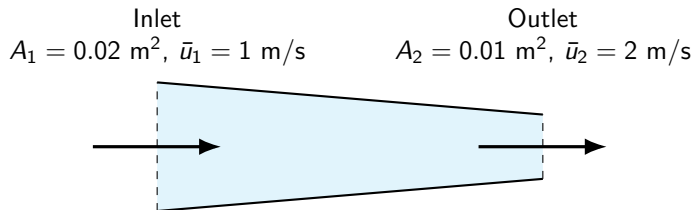
The amount inside the control volume remains unchanged because the entering and leaving rates are equal.

Equal entering and leaving rates do not mean that either rate is zero.

Water can flow continuously through a pipe while the mass of water inside the chosen pipe section stays constant.

A tapering pipe: volumetric flow rate

Volumetric flow rate is the volume of fluid passing through a cross-section per unit time. Its symbol is Q and its SI unit is m^3/s .



A denotes cross-sectional area (m^2), and $\bar{u}$ denotes mean velocity normal to the section (m/s).

For steady flow of constant-density water,

$$Q_{\text{in}} = A_1 \bar{u}_1 = 0.02 \text{ m}^3/\text{s}, \quad Q_{\text{out}} = A_2 \bar{u}_2 = 0.02 \text{ m}^3/\text{s}.$$

Thus 0.02 m^3 enters and leaves every second. The smaller section carries the same volume per second at a higher mean speed.

The same tapering pipe: mass flow rate

Mass flow rate is the mass of fluid passing through a cross-section per unit time. Its symbol is $\dot{m}$ and its SI unit is kg/s.

Water: density $\rho = 1000 \text{ kg/m}^3$. The volumetric flow rate at each section is $Q = 0.02 \text{ m}^3/\text{s}$.

$$\dot{m}_{\text{in}} = \rho Q_{\text{in}} = 1000(0.02) = 20 \text{ kg/s},$$

$$\dot{m}_{\text{out}} = \rho Q_{\text{out}} = 1000(0.02) = 20 \text{ kg/s}.$$

No water crosses the pipe wall, and there is no generation or consumption of total mass. Therefore,

$\text{mass accumulation} = 20 - 20 = 0 \text{ kg/s}.$

Every second, 20 kg of water enters and 20 kg leaves. **Flow continues, but the mass inside the pipe section does not change.**

When can the time derivative pass inside the integral?

For a fixed region V ,

$$\frac{d}{dt} \int_V C(\mathbf{x}, t) dV = \int_V \frac{\partial C}{\partial t}(\mathbf{x}, t) dV.$$

The left side differentiates the *total amount*. The right side adds the *local changes at fixed positions*. The region and its boundary must be independent of time. Also, the integrand must allow differentiation under the integral sign. A sufficient condition on a finite region is that C and $\partial C/\partial t$ are continuous near the time considered. The same step is not generally valid for a moving boundary.

A one-dimensional example shows the boundary terms

Let $a(t)$ and $b(t)$ be moving endpoints (m), $\dot{a}$ and $\dot{b}$ their velocities (m/s), and $C(x, t)$ amount per unit length.

$$\frac{d}{dt} \int_{a(t)}^{b(t)} C \, dx = \int_a^b \frac{\partial C}{\partial t} \, dx + C(b, t)\dot{b} - C(a, t)\dot{a}.$$

Even if C is constant in time at every point, extending the interval increases its total amount. The endpoint terms account for the newly included or removed region. If both endpoints are fixed, the last two terms vanish. This is why a fixed-region condition matters.

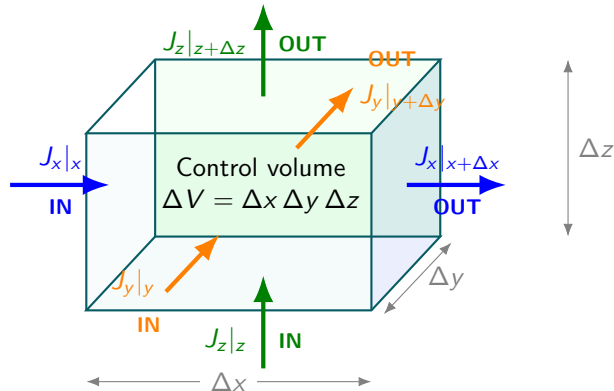
Moving boundaries: the volume transport identity

Let $V(t)$ be a moving region and $\mathbf{w}$ its boundary velocity (m/s). Then

$$\frac{d}{dt} \int_{V(t)} C dV = \int_{V(t)} \frac{\partial C}{\partial t} dV + \oint_{\partial V(t)} C \mathbf{w} \cdot \mathbf{n} dA.$$

The first term is the change at fixed positions. The second accounts for volume swept by the boundary. Only the normal boundary velocity affects this volume change. Tangential movement along the surface does not sweep additional volume. For a fixed boundary, $\mathbf{w} = \mathbf{0}$ and the second term is zero.

Derive the differential balance using a small box



A cube is a special case of this box. Side lengths Δx , Δy , Δz are in m; volume $\Delta V = \Delta x \Delta y \Delta z$ is in m^3 .

J_x , J_y , J_z are flux components. Arrows show positive coordinate directions. Negative flux reverses the actual flow.

Cube balance: amount entering and leaving the x faces

Observe the fixed box from time t to $t + \Delta t$, where Δt is an elapsed time (s). Each x face has area $\Delta y \Delta z$.

in through the lower x face = $\bar{J}_x|_x \Delta y \Delta z \Delta t$,

out through the upper x face = $\bar{J}_x|_{x+\Delta x} \Delta y \Delta z \Delta t$.

Therefore the signed amount entering minus leaving is

$$[\bar{J}_x|_x - \bar{J}_x|_{x+\Delta x}] \Delta y \Delta z \Delta t.$$

The bar means flux averaged over the face and time interval. This makes the finite-amount expression exact. For mass, each amount has unit kg.

Cube balance: add the y and z face pairs

The y faces each have area $\Delta x \Delta z$. The z faces each have area $\Delta x \Delta y$.

$$(\text{in} - \text{out})_y = [\bar{J}_y|_y - \bar{J}_y|_{y+\Delta y}] \Delta x \Delta z \Delta t,$$

$$(\text{in} - \text{out})_z = [\bar{J}_z|_z - \bar{J}_z|_{z+\Delta z}] \Delta x \Delta y \Delta t.$$

Add these to the x-face contribution. All six faces are now included.

These are signed contributions. The subtraction is valid even if an individual flux component is negative.

Cube balance: accumulation and net production

C is amount per volume; S is generation minus consumption per volume per time.

Let $\overline{C}(t)$ be the spatial average of C over the box at time t . Let $\overline{S}$ be the volume-and-time average over the interval.

$$\text{accumulated amount} = [\overline{C}(t + \Delta t) - \overline{C}(t)]\Delta V.$$

$$\text{generation} - \text{consumption} = \overline{S} \Delta V \Delta t.$$

For species mass, C has unit kg/m^3 and S has unit $\text{kg}/(\text{m}^3 \text{ s})$. Both expressions above therefore have unit kg .

Accumulation is positive for an increase and negative for a decrease.

Cube balance: accumulation equals in minus out plus source

Add the three face-pair contributions and the net production:

$$\begin{aligned} [\bar{C}(t + \Delta t) - \bar{C}(t)]\Delta V = & [\bar{J}_x|_x - \bar{J}_x|_{x+\Delta x}]\Delta y\Delta z\Delta t \\ & + [\bar{J}_y|_y - \bar{J}_y|_{y+\Delta y}]\Delta x\Delta z\Delta t \\ & + [\bar{J}_z|_z - \bar{J}_z|_{z+\Delta z}]\Delta x\Delta y\Delta t \\ & + \bar{S}\Delta V\Delta t. \end{aligned}$$

Every term is an amount. This is the finite control-volume balance before taking a limit.

The bar denotes the appropriate average, as defined on the preceding slides.

Cube balance: divide by volume and elapsed time

Use $\Delta V = \Delta x \Delta y \Delta z$ and divide each term by $\Delta V \Delta t$:

$$\begin{aligned} \frac{\bar{C}(t + \Delta t) - \bar{C}(t)}{\Delta t} = & - \frac{\bar{J}_x|_{x+\Delta x} - \bar{J}_x|_x}{\Delta x} \\ & - \frac{\bar{J}_y|_{y+\Delta y} - \bar{J}_y|_y}{\Delta y} \\ & - \frac{\bar{J}_z|_{z+\Delta z} - \bar{J}_z|_z}{\Delta z} + \bar{S}. \end{aligned}$$

For mass, every term now has unit $\text{kg}/(\text{m}^3 \text{ s})$. Increasing outward flux gives a negative contribution to accumulation.

Cube balance: take the limit in component form first

For smooth fields, let $\Delta x, \Delta y, \Delta z, \Delta t \rightarrow 0$. Each difference quotient tends to a partial derivative:

$$\frac{\partial C}{\partial t} = -\frac{\partial J_x}{\partial x} - \frac{\partial J_y}{\partial y} - \frac{\partial J_z}{\partial z} + S.$$

This is the local balance written explicitly in components.

Only now introduce the abbreviation

$$\nabla \cdot \mathbf{J} = \frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z}.$$

The same equation becomes $\partial C / \partial t = -\nabla \cdot \mathbf{J} + S$. The compact notation adds no new physical assumption.

Gauss's divergence theorem: the statement

For a vector field $\mathbf{J} = (J_x, J_y, J_z)$, the components J_x, J_y, J_z give flux in the x, y, z directions.

$$\oint_{\partial V} \mathbf{J} \cdot \mathbf{n} dA = \int_V \nabla \cdot \mathbf{J} dV.$$

The divergence is

$$\nabla \cdot \mathbf{J} = \frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z}.$$

It measures net outward transport per unit volume at a point. If $\mathbf{J}$ is mass flux, its divergence has unit $\text{kg}/(\text{m}^3 \text{ s})$. The theorem applies to a bounded region with a closed, piecewise smooth boundary and a sufficiently smooth field; continuously differentiable $\mathbf{J}$ is sufficient.

Why Gauss's theorem works

Divide the region into many small adjoining cells. Calculate signed outward transport through every cell face. At a shared face, one cell's outward normal is opposite to its neighbour's outward normal. The two contributions cancel if the field has the same value on that face. After adding all cells, only the outer surface contributions remain. The sum of the small-cell net outward transport is therefore the net outward transport through the whole boundary. In the small-cell limit, net outward transport divided by cell volume is divergence. Integrating it over the region gives the surface integral.

Divergence is not the flux itself

Consider $J_x = ax$, $J_y = J_z = 0$, on a box $0 \leq x \leq L$ of cross-sectional area A . Here L is length (m), A is area (m^2), and a has unit $\text{kg}/(\text{m}^3 \text{ s})$ for mass transport.

$$\text{surface net efflux} = [J_x(L) - J_x(0)]A = aLA.$$

$$\text{volume integral of divergence} = \int_V a \, dV = aLA.$$

Both calculations give the same kg/s . Flux is a rate per area; divergence is a rate per volume. A large but spatially uniform flux can have zero divergence.

From the integral balance to the local equation

For a fixed CV, move the time derivative inside and apply Gauss's theorem:

$$\int_V \left[\frac{\partial C}{\partial t} + \nabla \cdot \mathbf{J} - S \right] dV = 0.$$

This balance must hold for **every** sufficiently small region, not just one chosen region. If the bracket is continuous and nonzero at a point, a small neighbourhood has the same sign and a nonzero integral. That contradicts the balance. Therefore, at each smooth point,

$$\boxed{\frac{\partial C}{\partial t} + \nabla \cdot \mathbf{J} = S.}$$

From the local equation back to the integral balance

Start with $\partial C / \partial t + \nabla \cdot \mathbf{J} = S$ and integrate over any fixed region. Apply Gauss's theorem to the divergence term. Move the time derivative outside the volume integral using the fixed-region condition.

$$\frac{d}{dt} \int_V C dV + \oint_{\partial V} \mathbf{J} \cdot \mathbf{n} dA = \int_V S dV.$$

The two forms represent the same balance under the stated smoothness assumptions. One describes the whole region; the other describes each point. At discontinuities, retain the integral or weak form and impose the appropriate interface jump conditions. Do not differentiate a jump as if the field were smooth.

Exact finite amounts require integration or averages

During a finite interval $t_0 \leq t \leq t_1$, the fixed-CV balance is

$$B_{CV}(t_1) - B_{CV}(t_0) = - \int_{t_0}^{t_1} \oint_{\partial V} \mathbf{J} \cdot \mathbf{n} dA dt + \int_{t_0}^{t_1} \int_V S dV dt.$$

t_0 and t_1 are initial and final times (s). All terms are amounts. Flux times area times elapsed time is exact for a constant flux, or for the appropriate area-and-time average. Using one sampled point value for a varying field is an approximation. The same distinction applies to source rate times volume times time.

Reynolds transport theorem for a material region

Let $\mathcal{M}(t)$ contain the same particles, moving with fluid velocity $\mathbf{u}$ (m/s). Let $B_{sys} = \int_{\mathcal{M}(t)} C dV$ be their total property. The material boundary moves at $\mathbf{u}$, so the transport identity gives

$$\frac{dB_{sys}}{dt} = \int_{\mathcal{M}(t)} \frac{\partial C}{\partial t} dV + \oint_{\partial\mathcal{M}(t)} C \mathbf{u} \cdot \mathbf{n} dA.$$

This is a geometrical rate-of-change identity. It applies to mass, energy or momentum, but it does not say that each property is constant. We must supply the physical law separately: mass conservation, Newton's law, or the first law of thermodynamics.

Reynolds transport theorem using a moving CV

At the instant when a material system and a chosen CV coincide,

$$\frac{dB_{sys}}{dt} = \frac{d}{dt} \int_{V(t)} C dV + \oint_{\partial V(t)} C (\mathbf{u} - \mathbf{w}) \cdot \mathbf{n} dA.$$

$\mathbf{u}$ is fluid velocity and $\mathbf{w}$ boundary velocity, both in m/s. Their difference is fluid velocity relative to the boundary. The first term is change of the amount inside the moving CV. The second is net outward transport by matter crossing its moving boundary. For a fixed CV, $\mathbf{w} = \mathbf{0}$. For a material CV, $\mathbf{w} = \mathbf{u}$ on the boundary and the crossing term vanishes.

Derive the relative-velocity term rather than assume it

For coincident regions, the material derivative of the total property contains surface term $C\mathbf{u} \cdot \mathbf{n}$. The rate of the moving-CV integral contains surface term $C\mathbf{w} \cdot \mathbf{n}$. Subtract the second surface term from the first:

$$C\mathbf{u} \cdot \mathbf{n} - C\mathbf{w} \cdot \mathbf{n} = C(\mathbf{u} - \mathbf{w}) \cdot \mathbf{n}.$$

If the fluid and boundary move together, no particles cross. If a boundary moves into stationary fluid, its motion can still bring fluid into the chosen region. This is why absolute fluid velocity alone is insufficient for a moving CV.

Total mass: continuity in component form

For total mass, $C = \rho$, where ρ is density (kg/m^3). Relative total-mass flux is zero when velocity is the mass-average velocity. Chemical reactions do not produce total mass.

$$J_x = \rho u_x, \quad J_y = \rho u_y, \quad J_z = \rho u_z, \quad S = 0.$$

The component balance is

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_x)}{\partial x} + \frac{\partial(\rho u_y)}{\partial y} + \frac{\partial(\rho u_z)}{\partial z} = 0.$$

Each term has unit $\text{kg}/(\text{m}^3 \text{ s})$. For uniform constant density, divide by ρ to obtain $\partial_x u_x + \partial_y u_y + \partial_z u_z = 0$.

The next slide uses divergence to abbreviate the sum of the three spatial derivatives.

Apply mass conservation after applying RTT

For total mass, $\psi = 1$ and $C = \rho$. The mass of a material system is constant: $dB_{\text{sys}}/dt = 0$. For a moving CV,

$$\frac{d}{dt} \int_{V(t)} \rho dV + \oint_{\partial V(t)} \rho(\mathbf{u} - \mathbf{w}) \cdot \mathbf{n} dA = 0.$$

For a fixed CV, this becomes the usual integral continuity equation. Its local form is

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0.}$$

Zero on the right follows from *mass conservation*; it is not a consequence of RTT alone.

Mass following a particle and incompressibility

Define the material derivative $D/Dt = \partial/\partial t + \mathbf{u} \cdot \nabla$. It measures change while following a fluid particle. Expanding continuity gives

$$\boxed{\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{u} = 0.}$$

An incompressible flow has $D\rho/Dt = 0$, so $\nabla \cdot \mathbf{u} = 0$ where density is positive. Uniform constant density is a sufficient special case. A density field may be nonuniform but remain constant along particle paths. “Steady” and “incompressible” describe different conditions.

Bulk and relative transport: write each component first

Let u_x, u_y, u_z be fluid velocity components (m/s). Let j_x, j_y, j_z be flux components measured relative to the bulk fluid.

$$J_x = Cu_x + j_x, \quad J_y = Cu_y + j_y, \quad J_z = Cu_z + j_z.$$

Substitute these into the cube-derived balance:

$$\frac{\partial C}{\partial t} + \frac{\partial(Cu_x)}{\partial x} + \frac{\partial(Cu_y)}{\partial y} + \frac{\partial(Cu_z)}{\partial z} = -\frac{\partial j_x}{\partial x} - \frac{\partial j_y}{\partial y} - \frac{\partial j_z}{\partial z} + S.$$

Bulk motion and relative transport can both move the same property. The next slide writes this component equation in vector notation.

Bulk transport and transport relative to the bulk

For property per mass ψ , let $\mathbf{j}$ be its flux relative to the mass-average fluid velocity. Then total fixed-surface flux is

$$\mathbf{J} = C\mathbf{u} + \mathbf{j}, \quad C = \rho\psi.$$

$C\mathbf{u}$ is transport carried by bulk motion. $\mathbf{j}$ describes transport such as heat conduction or species diffusion. The moving-CV balance uses relative total flux $C(\mathbf{u} - \mathbf{w}) + \mathbf{j}$.

Conservation gives

$$\frac{\partial C}{\partial t} + \nabla \cdot (C\mathbf{u}) = -\nabla \cdot \mathbf{j} + S.$$

A transport law is needed to evaluate $\mathbf{j}$. Conservation alone does not determine it.

Species A: component fluxes and balance

w_A is species A mass fraction (dimensionless); ρw_A is species mass per volume (kg/m^3).
 $j_{A,x}$ is its diffusive flux in x ($\text{kg}/(\text{m}^2 \text{ s})$); define $j_{A,y}, j_{A,z}$ similarly.

$$J_{A,x} = \rho w_A u_x + j_{A,x} \quad \text{and similarly for } y, z.$$

Let r_A be reaction production minus consumption of A ($\text{kg}/(\text{m}^3 \text{ s})$).

$$\frac{\partial(\rho w_A)}{\partial t} = -\frac{\partial J_{A,x}}{\partial x} - \frac{\partial J_{A,y}}{\partial y} - \frac{\partial J_{A,z}}{\partial z} + r_A.$$

For a simple binary diffusion model, $j_{A,x} = -\rho D_A \partial w_A / \partial x$, where D_A is diffusivity (m^2/s).
Use corresponding expressions in y and z .

The next slide collects the three flux components into one vector.

Species balance: reactions conserve total mass

For species A, define w_A as mass fraction and $\rho_A = \rho w_A$ as species mass per volume (kg/m^3). Let $\mathbf{j}_A$ be diffusive mass flux ($\text{kg}/(\text{m}^2 \text{ s})$) and r_A net reaction production ($\text{kg}/(\text{m}^3 \text{ s})$).

$$\frac{\partial(\rho w_A)}{\partial t} + \nabla \cdot (\rho \mathbf{u} w_A + \mathbf{j}_A) = r_A.$$

For a simple binary mixture, one model is $\mathbf{j}_A = -\rho D_A \nabla w_A$, where D_A is diffusivity (m^2/s). For mass-average velocity, the species diffusive fluxes sum to zero. Reaction mass-production rates also sum to zero. Summing species balances therefore recovers total-mass continuity.

Species composition: mass fraction and mole fraction

For a binary mixture of species A and B, let m_A, m_B be masses (kg) and n_A, n_B amounts of substance (mol).

$$w_A = \frac{m_A}{m_A + m_B}, \quad y_A = \frac{n_A}{n_A + n_B}.$$

w_A is the mass fraction; y_A is the mole fraction. Both are dimensionless. A mole fraction of 0.20 means 20% of the moles are A.

Let M_A, M_B be molar masses (kg/mol), so $m_A = M_A n_A$.

$$w_A = \frac{M_A y_A}{M_A y_A + M_B (1 - y_A)}.$$

Mass fraction and mole fraction generally have different numerical values. They describe the same composition using different measures of amount.

Species concentration: amount per volume

For mixture volume V (m^3), define

$$c_A = \frac{n_A}{V}, \quad c_T = \frac{n_A + n_B}{V}, \quad c_A = c_T y_A.$$

c_A is species A molar concentration and c_T total molar concentration, both in mol/m^3 .

$$\rho_A = \frac{m_A}{V} = \rho w_A = M_A c_A.$$

ρ_A is species A mass concentration (kg/m^3); ρ is mixture mass density (kg/m^3).

A fraction describes the share of the mixture. A concentration describes the amount in a specified volume. Converting between them changes the units and numerical values, not the underlying material.

Fick's law: the same binary diffusivity in both forms

For the binary Fickian model used here, write $D = D_{AB}$ (m^2/s). This is the D_A used on the preceding species slides.

$$j_{A,x} = -\rho D \frac{\partial w_A}{\partial x}$$

$$N_{A,x}^d = -c_T D \frac{\partial y_A}{\partial x}.$$

$j_{A,x}$ is diffusive mass flux relative to the **mass-average** velocity, in $\text{kg}/(\text{m}^2 \text{ s})$.

$N_{A,x}^d$ is diffusive molar flux relative to the **molar-average** velocity, in $\text{mol}/(\text{m}^2 \text{ s})$. The superscript d denotes diffusion.

The same D occurs in both equations. Density or total molar concentration supplies the amount-per-volume factor. The flux units and reference velocities differ.

These are the ordinary binary composition-gradient laws; additional driving forces require their own terms. Welty, 5th ed., Chapter 24, Eqs. (24-16)–(24-17).

Concentration forms: D remains D

If total molar concentration c_T is spatially constant,

$$\frac{\partial c_A}{\partial x} = c_T \frac{\partial y_A}{\partial x}, \quad \boxed{N_{A,x}^d = -D \frac{\partial c_A}{\partial x}}.$$

If mixture mass density ρ is spatially constant,

$$\frac{\partial \rho_A}{\partial x} = \rho \frac{\partial w_A}{\partial x}, \quad \boxed{j_{A,x} = -D \frac{\partial \rho_A}{\partial x}}.$$

The prefactor is absorbed into the concentration gradient. D **keeps its value and its unit m^2/s .**

When c_T varies, $\partial_x c_A = c_T \partial_x y_A + y_A \partial_x c_T$. Thus one cannot simply replace $c_T \partial_x y_A$ by $\partial_x c_A$. The same condition applies to the mass form when ρ varies.

Example: two concentration descriptions give the same flux

Let $D = 10^{-5} \text{ m}^2/\text{s}$ and constant $c_T = 100 \text{ mol/m}^3$. Across 0.01 m , mole fraction y_A decreases linearly from 0.20 to 0.10 .

$$\frac{\partial y_A}{\partial x} = -10 \text{ m}^{-1}, \quad N_{A,x}^d = -100(10^{-5})(-10) = 0.01 \text{ mol}/(\text{m}^2 \text{ s}).$$

The corresponding concentrations are $c_A = 20$ and 10 mol/m^3 :

$$\frac{\partial c_A}{\partial x} = -1000 \text{ mol/m}^4, \quad N_{A,x}^d = -(10^{-5})(-1000) = 0.01 \text{ mol}/(\text{m}^2 \text{ s}).$$

Both descriptions give the same molar flux in $+x$. **We did not change D .** We changed the way composition was expressed.

Example continued: express the result as a mass flux

For this comparison, additionally take equal molar masses: $M_A = M_B = 0.020 \text{ kg/mol}$. Then $w_A = y_A$, $\rho = c_T M_A = 2 \text{ kg/m}^3$, and the mass- and molar-average velocities coincide.

$$j_{A,x} = -\rho D \frac{\partial w_A}{\partial x} = -2(10^{-5})(-10) = 2 \times 10^{-4} \text{ kg/(m}^2 \text{ s)}.$$

Equivalently, ρ_A decreases from 0.40 to 0.20 kg/m^3 across 0.01 m:

$$j_{A,x} = -D \frac{\partial \rho_A}{\partial x} = -(10^{-5})(-20) = 2 \times 10^{-4} \text{ kg/(m}^2 \text{ s)}.$$

Here $j_{A,x} = M_A N_{A,x}^d$. With unequal molar masses, the two average velocities generally differ, so diffusive fluxes must be converted with their reference velocities included.

The binary diffusivity remains $10^{-5} \text{ m}^2/\text{s}$ throughout this example.

Heat conduction: components before vector notation

Let T be temperature (K), k thermal conductivity (W/(m K)), and q_x, q_y, q_z heat-flux components (W/m²).

$$q_x = -k \frac{\partial T}{\partial x}, \quad q_y = -k \frac{\partial T}{\partial y}, \quad q_z = -k \frac{\partial T}{\partial z}.$$

For a stationary solid, let c be specific heat (J/(kg K)), ρ density (kg/m³), and s_T internal heating (W/m³). Then

$$\rho c \frac{\partial T}{\partial t} = -\frac{\partial q_x}{\partial x} - \frac{\partial q_y}{\partial y} - \frac{\partial q_z}{\partial z} + s_T.$$

Each term is energy transfer or accumulation per volume per time (W/m³). The next slide abbreviates these component laws using gradient and divergence.

Heat conduction: define the temperature and the heat flux

Let T be temperature (K), k thermal conductivity (W/(m K)), and $\mathbf{q}$ conductive heat flux (W/m²). Fourier's law for an isotropic material is

$$\mathbf{q} = -k \nabla T.$$

∇T is the temperature gradient (K/m) and points towards increasing temperature. The negative sign makes heat flow towards lower temperature. For a stationary solid, neglect mechanical work and let c be its appropriate specific heat capacity (J/(kg K)); commonly $c \simeq c_p$, where c_p is constant-pressure specific heat capacity with the same unit. Let s_T be volumetric heating (W/m³).

$$\rho c \frac{\partial T}{\partial t} = -\nabla \cdot \mathbf{q} + s_T.$$

Heat balance and thermal diffusivity

Substituting Fourier's law gives

$$\rho c \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + s_T.$$

For constant ρ, c, k and no source, define thermal diffusivity $\alpha = k/(\rho c)$ (m^2/s):

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T.$$

The Laplacian $\nabla^2 T$ is the sum of the three second spatial derivatives; its unit is K/m^2 . The temperature equation is a simplified energy equation. General flowing-fluid energy includes pressure work, mechanical work and other terms. It is not automatically a diffusion law for $\rho c_p T$.

Momentum: first choose the x component

Momentum in the x direction is mass times x velocity. Thus its amount per volume is $C = \rho u_x$, in $\text{kg}/(\text{m}^2 \text{ s})$.

Flow through an x face carries x -momentum at flux $\rho u_x u_x$. Flow through a y face carries the same momentum component at flux $\rho u_x u_y$. A z face carries flux $\rho u_x u_z$.

convective x -momentum fluxes: ρu_x^2 , $\rho u_x u_y$, $\rho u_x u_z$.

Each has unit N/m^2 . Transport direction and momentum direction are different: a y -directed flow can carry x -momentum.

Newton's law supplies the physical balance: momentum accumulation equals net momentum entering plus external force.

Momentum: define the force components on cube faces

Define σ_{ij} as the force in direction i per area on a face with outward normal in $+j$ (Pa = N/m²).

For x force, opposite-face pairs give the following net forces in the small-box limit:

$$F_x^{(x \text{ faces})} \simeq \frac{\partial \sigma_{xx}}{\partial x} \Delta V,$$

$$F_x^{(y \text{ faces})} \simeq \frac{\partial \sigma_{xy}}{\partial y} \Delta V,$$

$$F_x^{(z \text{ faces})} \simeq \frac{\partial \sigma_{xz}}{\partial z} \Delta V.$$

Body force adds $\rho b_x \Delta V$, where b_x is body acceleration in x (m/s²). For example, gravity contributes through the appropriate component.

Stress differences, not stress alone, determine the net surface force on the cube.

Momentum: the complete x-component equation

Divide the cube's momentum balance by its volume and take the smooth-field limit:

$$\frac{\partial(\rho u_x)}{\partial t} + \frac{\partial(\rho u_x u_x)}{\partial x} + \frac{\partial(\rho u_x u_y)}{\partial y} + \frac{\partial(\rho u_x u_z)}{\partial z} = \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z} + \rho b_x.$$

Left: accumulation plus net outward transport of x-momentum. Right: surface force plus body force per volume.

Let pressure be p (Pa) and viscous stress components be τ_{ij} (Pa). Since $\sigma_{xx} = -p + \tau_{xx}$, $\sigma_{xy} = \tau_{xy}$ and $\sigma_{xz} = \tau_{xz}$, the right side is

$$-\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} + \rho b_x.$$

Every term has unit N/m³.

Momentum: generalize the component, then use tensors

Repeat the same derivation for y - and z -momentum. For any one component $i = x, y, z$:

$$\frac{\partial(\rho u_i)}{\partial t} + \sum_{j=x,y,z} \frac{\partial(\rho u_i u_j)}{\partial x_j} = \sum_{j=x,y,z} \frac{\partial \sigma_{ij}}{\partial x_j} + \rho b_i.$$

Here x_j is the coordinate x , y or z . The sum adds transport or force contributions from all three face pairs.

Collecting the three momentum equations gives

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = \nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{b}.$$

The tensor $\mathbf{u} \otimes \mathbf{u}$ has component $u_i u_j$. Here tensor divergence means $(\nabla \cdot \boldsymbol{\sigma})_i = \sum_j \partial \sigma_{ij} / \partial x_j$.

$\boldsymbol{\sigma} = -p\mathbf{I} + \boldsymbol{\tau}$, with $\mathbf{I}$ the dimensionless identity tensor. Compact notation represents the same three component equations.

A simple viscous shear example

For unidirectional velocity $u_x(y, t)$, define dynamic viscosity μ (Pa s). In a Newtonian fluid,

$$\tau_{xy} = \mu \frac{\partial u_x}{\partial y}, \quad j_{p_x, y} = -\tau_{xy}.$$

τ_{xy} is the x -directed traction on a surface with normal $+y$. $j_{p_x, y}$ is outward transport of x -momentum in the $+y$ direction; both have unit Pa. For constant density and viscosity, no pressure gradient or body force, and no streamwise variation,

$$\frac{\partial u_x}{\partial t} = \nu \frac{\partial^2 u_x}{\partial y^2}, \quad \nu = \mu / \rho.$$

Kinematic viscosity ν has unit m^2/s . The general Newtonian law uses the velocity-gradient tensor, not a scalar potential gradient.

Similar diffusion equations require stated assumptions

The temperature, species and simple shear equations can have the form

$$\partial\phi/\partial t = \Gamma\nabla^2\phi.$$

ϕ is the selected field: temperature, mass fraction or one velocity component. Γ is its constant diffusivity, in m^2/s . The characteristic diffusion time is $t_d \sim L^2/\Gamma$, where L is distance (m) and t_d time (s). Define dimensionless Prandtl, Schmidt and Lewis numbers:

$$\text{Pr} = \nu/\alpha, \quad \text{Sc} = \nu/D_A, \quad \text{Le} = \alpha/D_A.$$

These compare diffusivities. The common mathematical form does not make the physical transport laws identical.

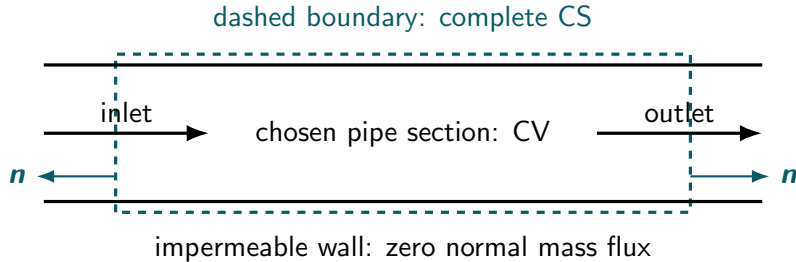
Why a gradient of density is not a universal driving force

The product rule gives

$$\nabla(\rho\psi) = \rho\nabla\psi + \psi\nabla\rho.$$

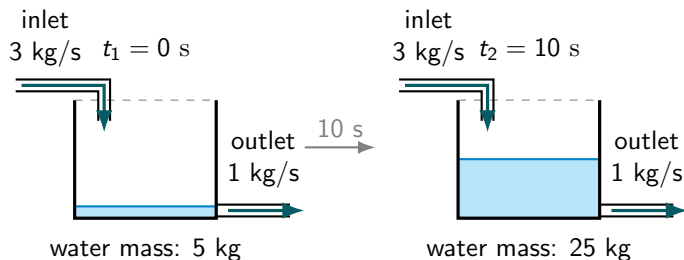
Thus $-\Gamma\nabla(\rho\psi)$ and $-\rho\Gamma\nabla\psi$ differ when $\psi\nabla\rho \neq 0$. A dimensionally correct expression is not automatically a valid physical law. Fourier conduction uses temperature gradient; simple Newtonian shear uses velocity gradient; species transport depends on the mixture and reference velocity. For constant c_p , $h = h_0 + c_p(T - T_0)$ defines specific enthalpy h (J/kg) relative to reference enthalpy h_0 (J/kg) and reference temperature T_0 (K). It does not make ρh a universal heat-diffusion field.

Diagram: the complete control surface



The outward normal points left at the inlet and right at the outlet. It always points out of the chosen region.

Example: A tank gains water



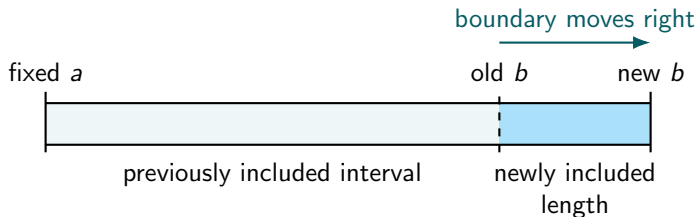
Choose the tank as the fixed control volume. In this illustration, its water capacity is 50 kg : it is 10% full at t_1 and half full at t_2 . Density is constant. The drawing is schematic.

$$\text{accumulation} = 3 - 1 = 2 \text{ kg/s}, \quad \text{mass gained in } 10 \text{ s} = 20 \text{ kg}.$$

The level rises because more water enters than leaves. Both pipes continue to carry water.

Example: A moving endpoint changes the amount

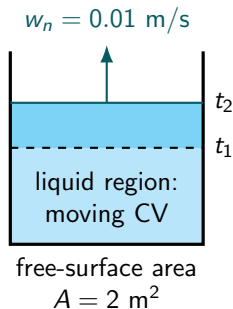
A line contains a uniform mass per length $C = 2$ kg/m. The left endpoint a is fixed; the right endpoint $b(t)$ moves at $\dot{b} = 0.5$ m/s.



The field does not change at fixed positions: $\partial C / \partial t = 0$. The moving endpoint nevertheless includes additional mass.

$$\frac{d}{dt} \int_a^{b(t)} C \, dx = C \dot{b} = 2(0.5) = 1 \text{ kg/s}.$$

Example: A rising surface adds volume to the chosen region



Choose the liquid alone as the control volume. Its free surface rises and its volume increases.

Density is $\rho = 1000 \text{ kg/m}^3$ and remains constant.

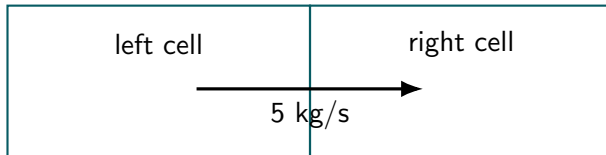
The local density-change term is zero.

$w_n = \mathbf{w} \cdot \mathbf{n}$ is the outward normal boundary speed (m/s).

$$\oint \rho \mathbf{w} \cdot \mathbf{n} dA = \rho w_n A$$
$$= 1000(0.01)(2) = 20 \text{ kg/s}.$$

The shaded upper strip is the volume added as the boundary rises. Its height is exaggerated for clarity.

Diagram: cancellation at the shared face



outward contribution:

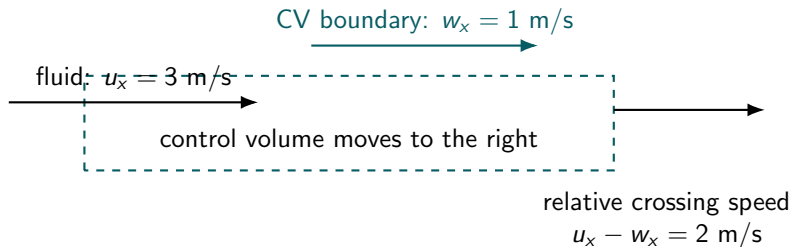
+5 kg/s

outward contribution:

-5 kg/s

The arrows describing the actual flow agree. The outward normals for the two cells are opposite. Adding their balances cancels this internal transfer.

Example: A translating CV measures relative flow



Density is $\rho = 1000 \text{ kg/m}^3$; each end has area $A = 0.01 \text{ m}^2$.

$$\text{mass crossing one end per second} = \rho(u_x - w_x)A = 20 \text{ kg/s}.$$

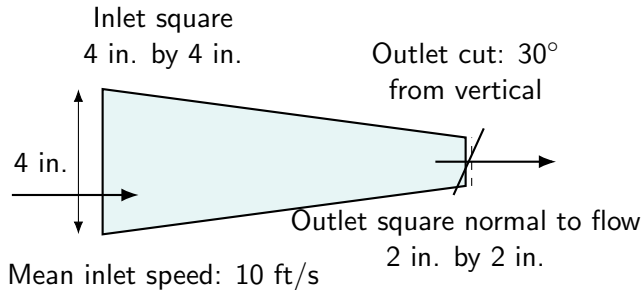
The fluid overtakes the moving boundary. In this uniform stream, 20 kg/s enters the moving CV and 20 kg/s leaves it. The mass inside stays constant.

Welty Problem 4.4: statement

“Water enters a 4-in. square channel as shown at a velocity of 10 fps. The channel converges to a 2-in. square configuration as shown at the discharge end. The outlet section is cut at 30° to the vertical as shown, but the mean velocity of the discharging water remains horizontal. Find the exiting water’s average velocity and total rate of flow.”

Source: Welty et al., 5th edition, Chapter 4, p. 39.

Problem 4.4: identify the two cross-sections



The fluid velocity is horizontal. The drawing is schematic; dimensions are given by the labels, not by its apparent heights. The slanted cut is not the area normal to flow.

Problem 4.4: define the sections and solve

A_1 and A_2 are areas perpendicular to the horizontal flow (ft^2). $\bar{u}_1$ and $\bar{u}_2$ are mean speeds (ft/s). Q is volumetric flow rate (ft^3/s). Convert inches to feet using $12 \text{ in.} = 1 \text{ ft}$.

$$A_1 = (4/12)^2 = 1/9, \quad A_2 = (2/12)^2 = 1/36.$$

Steady incompressible flow through impermeable walls has equal inlet and outlet volume flow rates:

$$\bar{u}_2 = \bar{u}_1 A_1 / A_2 = 10 \times 4 = 40 \text{ ft/s}.$$

$$Q = \bar{u}_1 A_1 = 10/9 = 1.11 \text{ ft}^3/\text{s}.$$

The smaller area requires a larger mean speed to carry the same volume per second.

Problem 4.4: an oblique surface gives the same flow

Let A_s be the actual slanted area and θ the angle between velocity and outward normal. For uniform speed u , the normal speed is $u \cos \theta$.

$$A_{\perp} = A_s \cos \theta, \quad Q = u \cos \theta A_s = u A_{\perp}.$$

$A_{\perp}$ is the projected area perpendicular to the velocity. In this figure, the outlet cut is 30° from vertical, so its normal is 30° from horizontal. The slanted area is larger, but the normal speed is smaller. The product stays the same. Use the *actual area with normal speed*, or the *projected area with full speed*; do not mix them.

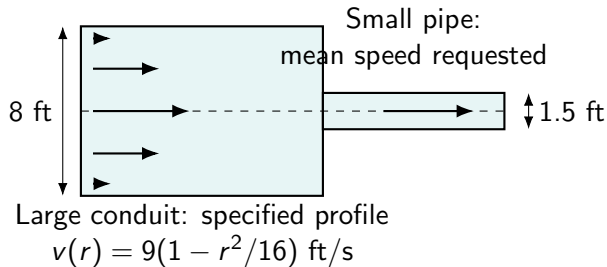
Welty Problem 4.3: statement

“Water is flowing through a large circular conduit with a velocity profile given by the equation $v = 9(1 - r^2/16)$ fps. What is the average water velocity in the 1.5-ft pipe?”

The book's figure labels the large conduit diameter as 8 ft and the downstream pipe diameter as 1.5 ft. Both dimensions are needed.

Source: Welty et al., 5th edition, Chapter 4, p. 39.

Problem 4.3: the profile belongs to the large conduit



r measures radial distance from the large-conduit axis. Integrate the full large-conduit flow first. Then divide by the smaller-pipe area. Diameters are labelled; the drawing is not to scale.

Problem 4.3: integrate over the large conduit

Let r be distance from the large-conduit centre (ft), $R = 4$ ft its radius, and $v(r)$ its axial speed (ft/s). In the supplied profile, 16 represents 16 ft^2 . A thin circular ring has area $dA = 2\pi r dr$. Its volume flow is $v(r)dA$. Add all rings:

$$Q = \int_0^4 9(1 - r^2/16)2\pi r dr = 72\pi \text{ ft}^3/\text{s}.$$

The average speed in the large conduit is $Q/(\pi R^2) = 4.5 \text{ ft/s}$. This is not yet the answer asked for: the problem asks about the smaller pipe.

Problem 4.3: transfer the flow to the smaller pipe

For steady incompressible flow, the same Q passes through both sections. The smaller-pipe radius is $R_p = 0.75$ ft and its area is $A_p = \pi R_p^2$.

$$\bar{v}_p = Q/A_p = \frac{72\pi}{\pi(0.75)^2} = 128 \text{ ft/s.}$$

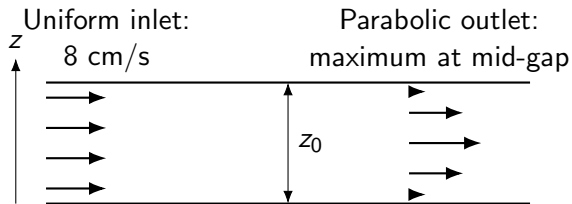
$\bar{v}_p$ is the mean speed in the smaller pipe. Integrating the specified profile only from 0 to 0.75 ft would use the central part of the upstream conduit, not its entire flow. That calculation does not answer this problem. This result follows from the prescribed flow and continuity; it does not predict the pressure needed to drive it.

Welty Problem 4.22: statement

“Incompressible steady flow in the inlet between parallel plates is uniform, $V_0 = 8$ cm/s, while downstream, the flow develops into the parabolic profile $v_x = az(z_0 - z)$, where a is a constant. What is the maximum value of v_x ?”

Source: Welty et al., 5th edition, Chapter 4, p. 42.

Problem 4.22: the gap stays the same



z is transverse distance from the lower plate; z_0 is plate separation. Both are lengths. The inlet and outlet widths into the page are equal. Arrow lengths indicate relative speeds, not distances travelled.

Problem 4.22: mean speed and maximum speed

z is distance across the gap (cm), z_0 gap width (cm), and a a profile coefficient ($1/(\text{cm s})$). v_x is axial speed (cm/s). The gap and width are unchanged, so continuity requires downstream mean speed $\bar{v} = V_0 = 8 \text{ cm/s}$.

$$\bar{v} = \frac{1}{z_0} \int_0^{z_0} az(z_0 - z) dz = \frac{az_0^2}{6}.$$

The maximum occurs at $z = z_0/2$, giving $v_{\max} = az_0^2/4$.

$$\bar{v} = \frac{2}{3} v_{\max} \quad \Rightarrow \quad \boxed{v_{\max} = 12 \text{ cm/s.}}$$

Conservation fixes the total flow. The specified profile determines how that flow is distributed across the gap.

Welty Problem 4.16: statement

“The V-shaped tank has width w into the paper and is filled from the inlet pipe at volume flow Q . Derive expressions for (a) the rate of change dh/dt and (b) the time required for the surface to rise from h_1 to h_2 . ”

The figure measures each 20° wall angle from the horizontal. The liquid height h is vertical from the bottom of the V.

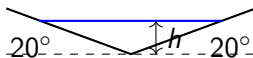
Source: Welty et al., 5th edition, Chapter 4, p. 41.

Problem 4.16: define the geometry before differentiating

Let $\theta = 20^\circ$, h liquid height (m), and w constant tank width into the page (m). The inlet volume flow rate Q is constant (m^3/s); there is no outlet. The half-width at the free surface is $h/\tan \theta$. The triangular filled area is

$$\frac{1}{2}(2h/\tan \theta)h = h^2 \cot \theta.$$

Therefore the liquid volume is $V_\ell = wh^2 \cot \theta$ (m^3).



Problem 4.16: the level rises at a changing rate

With constant density and no outlet, added volume per second equals Q :

$$\frac{dV_\ell}{dt} = Q = 2wh \cot \theta \frac{dh}{dt}.$$

$$\boxed{\frac{dh}{dt} = \frac{Q \tan \theta}{2wh}}.$$

The filling time from initial height h_1 to final height h_2 is

$$\boxed{t_{1 \rightarrow 2} = \frac{w \cot \theta}{Q} (h_2^2 - h_1^2)}.$$

The tank becomes wider at greater height. The same incoming volume therefore produces a smaller height increase per second. The height-rate formula applies for $h > 0$; use the volume relation at the empty starting point.

Welty Problem 4.7: statement

“Salt water containing 1.92 lb/gal of salt flows at a fixed rate of 2 gal/min into a 100-gal tank, initially filled with fresh water. The density of the incoming solution is 71.8 lb/ft³. The solution, kept uniform by stirring, flows out at a fixed rate of 19.2 lb/min.”

Source: Welty et al., 5th edition, Chapter 4, p. 40.

Welty Problem 4.7: the three questions

- “(a) How many pounds of salt will there be in the tank at the end of 1 h and 40 min?”*
- “(b) What is the upper limit for the number of pounds of salt in the tank if the process continues indefinitely?”*
- “(c) How much time will elapse while the quantity of salt in the tank changes from 100 to 150 lb?”*

Source: Welty et al., 5th edition, Chapter 4, p. 40. The quotation is followed by the assumptions used for the calculation.

Problem 4.7: state the density approximation

Let M be solution mass (lb), S_s salt mass (lb), t time (min), and $\dot{m}$ solution mass flow rate (lb/min). Using the US gallon conversion, 71.8 lb/ft^3 is approximately 9.60 lb/gal . We use the rounded, constant-mixture-density model: $\dot{m}_{in} = \dot{m}_{out} = 19.2 \text{ lb/min}$ and $M = 960 \text{ lb}$ in 100 gal . This also neglects the density difference between fresh water and brine. Without a density–composition model, an exact variable-density filling calculation is not specified. This approximation keeps volume and mass constant and makes the indefinite limit meaningful. A small mismatch caused by conversion rounding must not be extrapolated indefinitely.

Problem 4.7: salt entering and leaving

Salt enters at $(1.92 \text{ lb/gal})(2 \text{ gal/min}) = 3.84 \text{ lb/min}$. Stirring makes the outlet salt mass fraction equal to S_s/M .

$$\frac{dS_s}{dt} = 3.84 - 19.2 \frac{S_s}{960} = 3.84 - 0.02S_s, \quad S_s(0) = 0.$$

The initial fresh water contains no salt. Solving this first-order equation gives

$$S_s(t) = 192(1 - e^{-0.02t}) \text{ lb}, \quad t \text{ in min.}$$

The factor 0.02 has unit min^{-1} . At 100 min , $S_s = 166.0 \text{ lb}$. As $t \rightarrow \infty$, $S_s \rightarrow 192 \text{ lb}$. The net salt accumulation becomes smaller as outlet salt concentration approaches inlet concentration.

Problem 4.7: time to change from 100 to 150 lb

For any salt mass $S_s < 192$ lb,

$$t(S_s) = -50 \ln(1 - S_s/192) \text{ min.}$$

Subtract the two times:

$$\Delta t = 50 \ln \left(\frac{192 - 100}{192 - 150} \right) = 39.2 \text{ min.}$$

Δt is elapsed time between those salt masses, not time from the initially fresh tank. All three results use the explicitly stated constant-density, well-mixed approximation. The conservation law and the mixing assumption are separate: conservation gives the salt balance; mixing gives the outlet composition.

Checks to make in every problem

1. State the CV, its boundary and whether either moves.
2. Name every quantity before using its symbol and unit.
3. State the physical law, then evaluate each balance term.
4. Use mean velocities or integrate nonuniform profiles.
5. Check signs: outward normal transport is positive.
6. Check dimensions and consistency of the given values.
7. Explain whether the result meets the stated assumptions.

For mass use conservation; for momentum include forces; for energy include heat and work. RTT connects system and CV descriptions but does not replace these physical laws.

References and source notes

Welty, Wicks, Wilson and Rorrer, *Fundamentals of Momentum, Heat, and Mass Transfer*, 5th edition, Wiley: Chapter 4 (mass), Chapter 5 (momentum), Chapter 6 (energy).

Bird, Stewart and Lightfoot, *Transport Phenomena*, 2nd edition, Wiley.